Mean–Variance Optimization of Ten ETFs

Assignment 1: empirical analysis and chronological robustness

Wei Qi 226010267@link.cuhk.edu.cn

1. Data and estimation

The universe spans US equity (SPY, QQQ, IWM), international equity (EFA, EEM), aggregate bonds (AGG), long Treasuries (TLT), high-yield credit (HYG), gold (GLD), and US real estate (VNQ). The supplied Yahoo Finance adjusted-close snapshot covers 2021-10-11–2026-10-08, with 1254 prices and 1253 daily return observations per ETF. There are no missing prices in the analysis panel. Adjustments account for splits and distributions; USD prices describe US-listed instruments, not exclusively US economic exposures.

For daily log returns $\ell_{it} = \log(P_{it}/P_{i,t-1})$, the estimates are

$$\widehat{\mu} = m\bar{\ell}, \qquad \widehat{\Sigma} = m\frac{\sum_t(\ell_t - \bar{\ell})(\ell_t - \bar{\ell})^\top}{T-1}, \qquad m = 252.$$

The annualized mean standard error under independent observations is $ms_i/\sqrt{T}$, not annualized volatility divided by $\sqrt{T}$. Dependence would require a HAC estimator or block bootstrap. The table reports log means and volatilities; $e^{\widehat{\mu_i}} - 1$ is historical annualized geometric growth, not an expected arithmetic return.

Table 1: Annualized ETF log-return estimates.

ETF	Log mean (%)	Vol. (%)	Mean SE (%)
SPY	12.91	17.12	7.68
QQQ	15.37	22.95	10.29
IWM	5.81	22.35	10.02
EFA	8.71	16.60	7.45
EEM	7.62	19.98	8.96
AGG	-0.44	6.14	2.75
TLT	-8.45	15.68	7.03
HYG	3.36	7.54	3.38
GLD	16.83	19.00	8.52
VNQ	0.91	18.86	8.46

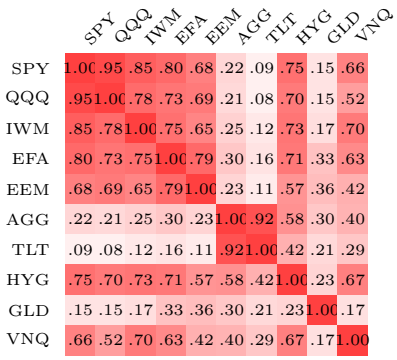


Figure 1: Daily log-return correlations.

2. Optimization and economic meaning

For a fully invested portfolio, solve

$$\min_w \tfrac{1}{2} w^\top \widehat{\Sigma} w \quad \text{subject to } \mathbf{1}^\top w = 1, \quad \widehat{\mu}^\top w = r,$$

with $w_i \geq 0$ added for the no-short-sale case. The minimum-variance portfolio (MVP) omits the target-return constraint. Efficient points lie above the MVP in expected return. Tangency maximizes $(\widehat{\mu}^\top w - r_f)/\sqrt{w^\top \widehat{\Sigma} w}$ with the assignment's annual hurdle $r_f = 3\%$.

This is a *log-moment proxy*: weighted asset log returns are not exact portfolio log returns. It abstracts from trading costs, borrow availability and financing. Short sales enlarge the feasible set, so the unrestricted frontier weakly dominates the constrained frontier at common feasible targets; this is an estimation result, not a guaranteed future return.

3. Full-sample efficient frontiers

Let $Q = \widehat{\Sigma}^{-1}$, $A = \mathbf{1}^\top Q \mathbf{1}$, $B = \mathbf{1}^\top Q \widehat{\mu}$, $C = \widehat{\mu}^\top Q \widehat{\mu}$, and $D = AC - B^2$. Then

$$w_{\text{MVP}} = Q\mathbf{1}/A, \qquad \sigma^2(r) = (Ar^2 - 2Br + C)/D.$$

The target-return weights are $Q(\lambda\mathbf{1} + \gamma\widehat{\mu})$, with $\left[\begin{smallmatrix} A & B \\ B & C \end{smallmatrix}\right](\lambda, \gamma)^\top = (1, r)^\top$. Implementation uses linear solves rather than an explicit inverse. Constrained weights are computed numerically and checked for budget, target and bound residuals.

If $d = \mathbf{1}^\top Q(\widehat{\mu} - r_f\mathbf{1}) > 0$, the unrestricted tangency is $w_T = Q(\widehat{\mu} - r_f\mathbf{1})/d$. Its capital allocation line is $r = r_f + S_T\sigma$. Cash has zero volatility; positions beyond the risky tangency point imply borrowing and leverage. Here the full-sample normalizer is 2.591.

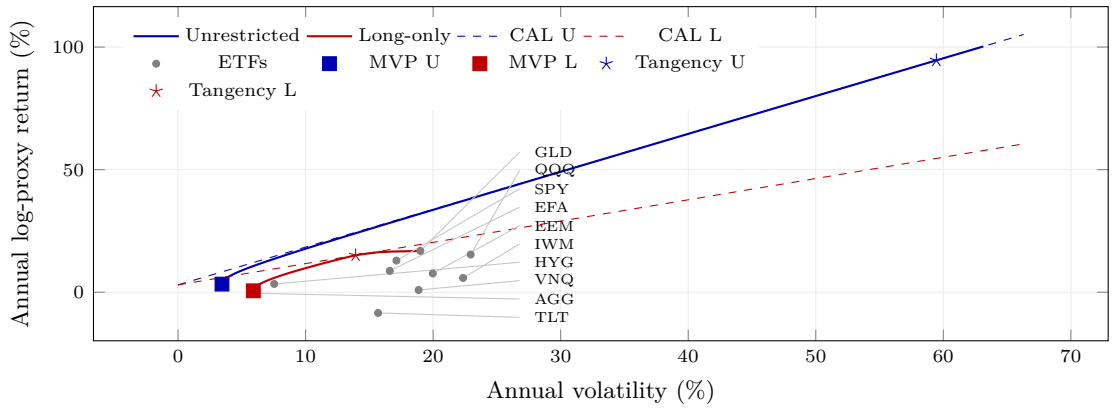


Figure 2: Both efficient frontiers, individual ETFs, MVPs, tangencies and CALs on one risk–return chart.

Table 2: Full-sample portfolio risk, return and leverage.

Portfolio	Log mean (%)	Vol. (%)	Proxy SR	Gross exposure
MVP U	3.31	3.45	0.089	2.55
MVP L	0.57	5.90	-0.412	1.00
Tangency U	94.59	59.46	1.540	16.93
Tangency L	15.08	13.92	0.868	1.00

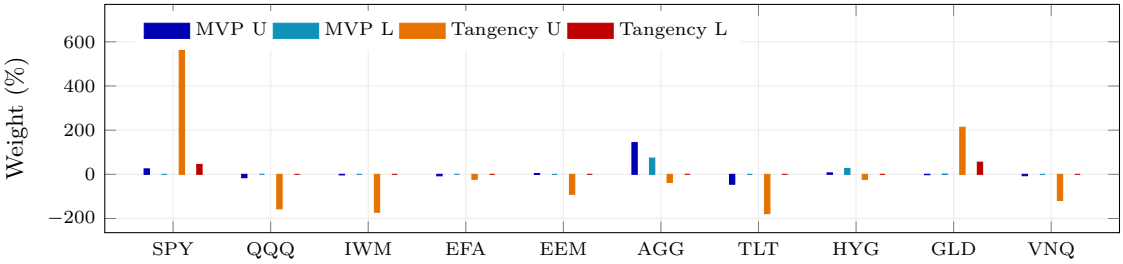


Figure 3: Full-sample portfolio allocations. Negative weights denote short positions.

The unrestricted tangency has 94.59% fitted log-proxy return, 59.46% volatility and gross exposure 16.93 times equity. Its largest position is 683.31% in SPY. These weights require substantial shorting and are not an investable recommendation. The long-only tangency instead combines 44.75% SPY and 55.25% GLD, with proxy Sharpe 0.868. The long-only MVP has volatility 5.90%, versus 3.45% unrestricted.

A high fitted Sharpe does not establish a persistent opportunity. Estimated mean differences are amplified by correlated assets and short positions. The MVP does not depend on mean estimates, but still relies on a stable covariance matrix. No-short-sale restrictions can reduce leverage and improve implementability while increasing the attainable variance at a given target return.

4. Out-of-sample robustness

Estimate weights only on the first three years, ending 2024-10-11, then hold those target weights fixed throughout the remaining two years. The split produces 755 in-sample and 498 out-of-sample returns with no shared return dates. Daily rebalancing uses exact portfolio simple returns $R_{p,t} = w^\top[\exp(\ell_t) - \mathbf{1}]$ and wealth $V_t = \prod_{s \le t}(1 + R_{p,s})$, beginning at $V_0 = 1$. Any daily return at or below -1 triggers insolvency handling rather than an invalid logarithm. The realized Sharpe uses daily simple excess returns over $r_{f,d} = (1.03)^{1/252} - 1$:

$$S = \sqrt{252}\,\frac{\overline{R_p - r_{f,d}}}{s(R_p - r_{f,d})}.$$

Reported geometric growth is $V_T^{252/T} - 1$. It is not the same statistic as the log-moment model's linear predicted return. On the training sample, $d = -2.475 < 0$. Thus there is no finite fully invested unrestricted Sharpe maximum. Its supremum is $\sqrt{C - B^2/A} = 1.901$, approached only with unbounded leverage; the normalized closed form is a Sharpe minimum, not a tangency maximum. The extra bounded diagnostic imposes $|w_i| \le 3$ and solves a different problem. It is labeled separately and does not replace the unavailable unrestricted optimum.

Table 3: In-sample and out-of-sample performance.

Strategy	IS SR	OOS SR	Growth (%)	Vol. (%)	Max DD (%)
MVP U	-0.263	0.154	3.38	2.61	-1.73
MVP L	-0.437	-0.027	2.78	4.30	-4.15
Tangency L	0.917	1.052	23.19	18.68	-17.50
Equal weight	0.153	0.796	12.21	11.60	-10.54
Bounded diagnostic	1.797	1.060	41.07	35.73	-46.32

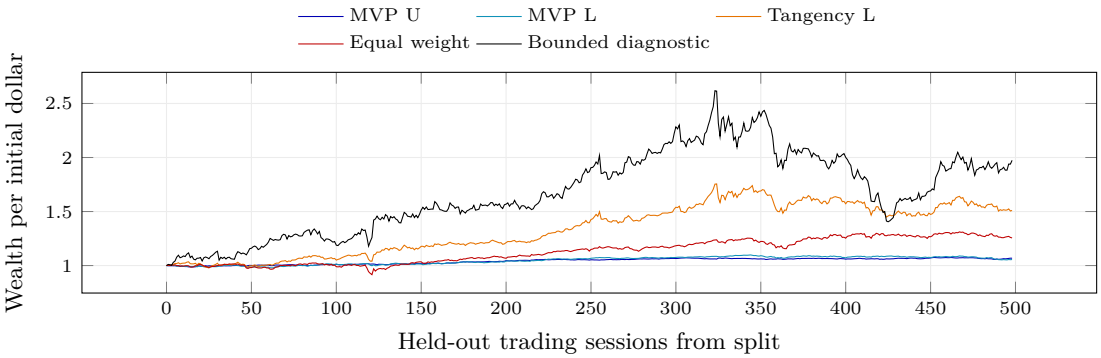


Figure 4: Out-of-sample wealth from fixed in-sample target weights, rebalanced daily from initial capital of one dollar.

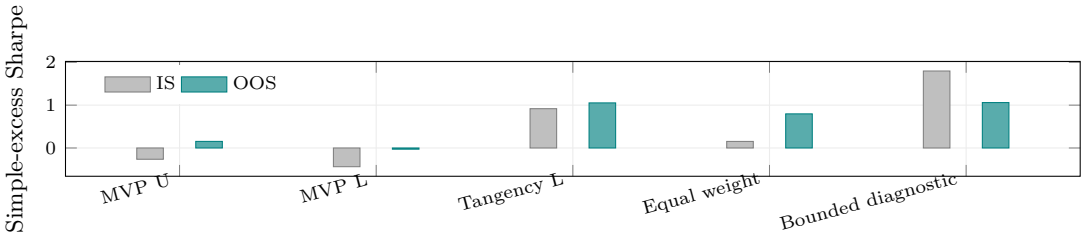


Figure 5: In-sample versus held-out Sharpe, computed using the same simple-return convention in both periods.

The long-only tangency earns Sharpe 1.052 versus 0.796 for equal weight, with total gains of 51.00% and 25.57%, respectively. Its maximum drawdown is -17.50% versus -10.54%. The bounded diagnostic earns 1.060 Sharpe but draws down -46.32%. Both MVPs reduce volatility but underperform the benchmark in Sharpe. Higher realized return therefore does not imply safer implementation.

5. Limitations and conclusion

Estimation and stability. Historical means are imprecise, and highly correlated ETFs make optimized long–short allocations particularly sensitive to small changes in estimated returns. The full-sample frontier is descriptive, not a forecast: its weights use observations unavailable at the training cutoff. Covariance is also time-varying, so even an MVP fitted without mean estimates need not retain its predicted risk. The selected universe is not a random sample of all investable assets; conclusions are conditional on these ETFs and this window.

Model and implementation. Mean–variance analysis treats volatility as the risk criterion and does not separately penalize tail losses, illiquidity or margin calls. Weighted log means approximate, rather than equal, the growth of rebalanced wealth. Daily rebalancing ignores spreads, fees, taxes and turnover; leveraged portfolios additionally require financing and short-borrow assumptions. A per-position cap does not bound gross exposure, and a solvent frictionless wealth path does not establish that a strategy could meet real margin requirements. The severe drawdown of the bounded diagnostic illustrates why return and Sharpe should not be read in isolation.

Evidence and practical conclusion. The held-out comparison shows that the long-only tangency beats equal weight in Sharpe in this interval, but also incurs a deeper drawdown and concentrates in two assets. The MVPs achieve low realized volatility without superior Sharpe. Thus the results support a risk–return trade-off, not universal dominance of optimization over diversification. One chronological split has substantial sampling uncertainty and provides neither a confidence interval nor evidence that the ranking will persist.

A stronger assessment would use repeated rolling training/test windows, block-bootstrap uncertainty estimates, covariance shrinkage and mean regularization. Gross-leverage and turnover constraints, realistic borrowing costs and transaction-cost stress tests should then be imposed before considering implementation. Equal weight remains a transparent estimation-free benchmark; a constrained portfolio is justified only when its prospective risk budget and net-of-cost performance offer a defensible advantage.